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CHEMOTHERAPY AND MANAGEMENT OF PANCREATIC CANCER

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ABSTRACT:

Pancreatic cancer remains one of the most fatal cancers in the world, and is often difficult to treat because the disease runs without symptoms a long time, is biologically aggressive, and responds poorly to available treatments. Although there has been notable progress in the development of molecular diagnostics and imaging technologies, timely detection of disease remains a major clinical challenge, and has a significant impact on prognosis and is associated with continued high mortality. This study has provided a thorough overview of the epidemiological trends, predisposing risk factors, pathophysiological mechanisms and key molecular alterations involved in pancreatic cancer progression and has been focused on oncogenic mutations in KRAS, TP53, SMAD4 and p16/CDKN2A. Immunological complexity of the disease is discussed in terms of the tumor microenvironment (stromal remodeling, desmoplasia, immunosuppressive cells and T-cell exhaustion as obstacles for effective immune surveillance). Diagnostic modalities are compared including computed tomography (CT), magnetic resonance imaging (MRI), positron emission tomography (PET), and endoscopic ultrasound (EUS) in terms of their diagnostic sensitivity and clinical applicability. The current standard-of-care therapies such as surgical resection, systemic chemotherapy, radiation therapy, and palliative therapies are also discussed critically, as well as their limitations, such as drug resistance mechanisms and impaired drug penetration in the dense tumor stroma. While the backbone of systemic treatment is based on gemcitabine, the results in terms of overall survival have not improved significantly in the last decades, strongly suggesting an urgent need for more effective therapeutic approaches. It is important, therefore, to understand the molecular, metabolic and immunological aspects of pancreatic cancer in order to develop new diagnostic strategies and precision-targeted treatments that will improve patient prognosis.as

Keywords: *Pancreatic cancer, Tumour microenvironment, Genetic mutations, Targeted therapy*

I. Introduction

Based on estimates, pancreatic cancer is responsible for 30,000 fatalities every year in the United States; this figure has stayed relatively constant over the past three to five years.[1] Because of the absence of effective therapies and the difficulty in detecting the illness at an early stage, pancreatic cancer remains one of the most challenging human cancers to treat. For the last 50 years, conventional therapies such as radiation, chemotherapy, surgery, or a combination of these have not been very effective on the evolution of this aggressive neoplasm. Therefore, a thorough understanding of its molecular biology is the only means available to successfully identify, prevent and treat pancreatic cancer.

Almost every patient with the disease forms metastases and ultimately dies, as a result of the serious metabolic consequences of uncontrolled tumour growth and untreated metastatic pancreatic cancer. The incidence and mortality rates are similar for pancreatic cancer, with the median age of diagnosis in the elderly, with only 10% of patients developing the disease at a younger age. Smoking is the major environmental risk factor: The incidence of pancreatic cancer rises sharply with age (from 9.8/100,000 in the 50-54 year age group to 57/100,000 in 70-75 year age group) according to data from the United States. In Korean men, the prevalence of smoking is estimated to be the highest in the world in 2001 (65.4%);[1517] this might account for the high pancreatic cancer mortality rate in this area. This paper summarizes recent developments in the epidemiology, risk factors, pathology, diagnosis and treatment of pancreatic cancer.

II. Epidemiology and risk factors

The American Cancer Society estimates that about 49,000 people in the U.S. will be diagnosed with pancreatic cancer and 41,000 will die from the disease in 2015. Pancreatic cancer is the fourth leading cause of cancer death in the U.S. In the world, about 200,000 people die of pancreatic cancer annually. The mortality rate linked to pancreatic cancer is continuously rising, and it is projected to become the second primary cause of cancer-related deaths in the US by the year 2030.[6] An escalation in pancreatic cancer mortality has also been documented within European populations, indicating the worldwide impact of the disease.[7] Numerous studies have explored the risk factors associated with pancreatic cancer. The top two risk factors cited are advancing age and smoking. Second, smoking cigarettes is thought to account for 25-29% of all cases of pancreatic cancer, with odds ratios ranging from 1.6 to 5.4.[9] Also, there are some inherited syndromes that have been linked to an increased risk of pancreatic cancer, such as hereditary pancreatitis, hereditary non-polyposis colorectal cancer, ataxia telangiectasia, Peutz-Jeghers syndrome, familial breast cancer, and familial atypical multiple-mole melanoma.[10,11]

III. Pathophysiology

The rapid acquisition of knowledge about the molecular biology of the disease has resulted in a significant advance in understanding the pathogenesis of pancreatic cancer. Like many other deadly diseases, pancreatic ductal melanoma is developing as a result of a series of acquired mutations. It is suspected that the mutations in the associated genes are expected to occur in a certain order in time, the majority of pancreatic ductal carcinomas being multigenic.[12, 13] A stepwise model of pancreatic ductal carcinogenesis was developed using histological and molecular inheritable biographies. Advancements in the understanding of pancreatic carcinogenesis have come with the discovery of a large number of familial pancreatic tumors that progress through an orderly series of events known as pancreatic intraepithelial neoplasia (PanIN).[14] Sometimes this pattern occurs in the most aggressive pancreatic tumors. Since the discovery of the first important inherited change, a mutation in the K-ras oncogene, the knowledge of the genetics of pancreatic cancer has grown rapidly. Over 85 pancreases In the very earliest stages of pancreatic cancer progression, ductal cancers have a cranky point mutation in the K-ras gene. A strategy for the early detection of pancreatic cancer involves screening for K-ras mutations within duodenal juice, pancreatic juice, and coprolite.[16, 17] The p16 tumor-suppressor gene is deactivated in roughly 95% of all pancreatic cancers,[18, 19] and its involvement has recently grown more common in pancreatic carcinogenesis.[20] The second most frequently inactivated tumor suppressor gene is TP53,

a well-known tumor suppressor gene on chromosome 17p. This gene's deactivation occurs in the late phases of tumorigenesis. The MADH4 gene (DPC4 or SMAD4) is inactivated in 55% of pancreatic adenocarcinomas.[21] Similar to TP53, the inactivation of MADH4 happens later in the development of pancreatic tumors. There are other, more rare, genetic defects linked to pancreatic cancer. All 42 pancreatic ductal cancers studied by Rosenblum et al.[27] had mutations in the K-ras oncogene. Mutational frequencies have been reported for the tumour suppressor genes p16 (82), TP53 (76), MADH4 (53) and BRCA2 (10).[28]

IV. Molecular biology of pancreatic cancer

These molecular disruptions give pancreatic cancer cells a major survival and growth edge, as evidenced by the development of phenotypes that are invasive and metastatic and resistant to all conventional treatment methods. Many growth factors and their receptors, such as members of the fibroblast growth factor family, [31] vascular endothelial growth factor [29, 30] and the epidermal growth factor family [28] have been identified to be overexpressed in deadly pancreatic cancer. In addition, several cytokines including interleukin 1, interleukin 6, various growth factors and their receptors like fibroblast growth factor (FGF), vascular

endothelial growth factor (VEGF) and epidermal growth factor (EGF) family are found to be up-regulated in lethal pancreatic cancer. Moreover, a variety of cytokines, such as interleukin 1, interleukin 6, The overgrowth of growth-promoting hormones, plus the absence of growth-inhibitory factors, is solely responsible for angiogenesis, metastasis, tone-adequate growth signals and the illusion of programmed cell death. The expression of these cytokines has in the past been associated with some processes.

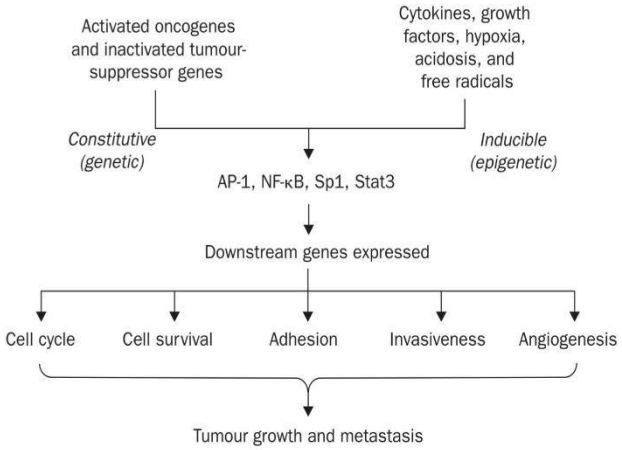
For example, vascular endothelial growth factor expression is strongly regulated by hypoxia and most solid tumors, including pancreatic cancers, are hypoxic. Low extracellular pH (acidosis), which occurs constantly within the growing tumor mass, particularly near the necrotic parts of the tumors, can result in vascular endothelial growth factor expression being significantly over-regulated, as shown by Shi and associates[37]. Moreover, they showed the activation of the gene for the substance interleukin-8 by acidity. The precursors of the angiogenic moles for pancreatic cancer are vascular endothelial growth factor and interleukin 8.

In a wide range of tumor types, an increase in the production of vascular endothelial growth factor is always observed in cells of the extreme periphery of the tumor where there is no apparent hypoxia or acidosis. Together, these observations support the idea that VEGF expression can be regulated by exogenous factors such as hormones, cytokines, and growth factors, and that, in turn, can have an impact on angiogenesis.[39]

Multiple implicit recap factor list spots, including HIF-1, AP-1, AP-2, Egr-1, Sp1, and many others have been identified in the course of vascular endothelial growth factor promoter research,[41] suggesting that more than one signal transduction pathway is involved in the regulation of vascular endothelial growth factor's recapitalization.

All of these likely play a role in the aggressive growth and drug resistance of pancreatic adenocarcinomas, for example, discriminatory activation of native SP1 is responsible for varying levels of vascular endothelial growth factor expression, and a constitutively activated launch directly affects native vascular endothelial growth factor expression in dead pancreatic cancer cells, as well as producing interleukin 8 in the absence of apparent external stimuli. The current focus of pancreatic cancer research is to understand the upstream molecular pathways that trigger the native activation of these recap factors.

Fig 1: Molecular biology of pancreatic tumor growth and metastasis is shown.



Proteins that are involved in several metastases, such as interleukin 8 and vascular endothelial growth factor, may exhibit non-normal expression due to alterations in various recap-factor conditionings. Two possible ways by which these adaptations are interpreted are proposed (Figure 1). The first mechanism is inherited mutations in oncogenes, suppressor genes or both (K-ras, TP53) that cause the recapitalization factors to be inherently activated. This impact might be especially evident during the initial phases of pancreatic cancer. Numerous recap factors are also activated during the later phase of the pancreatic cancer progression process, and these trigger the upregulation of proteins involved in metastasis in the context of significant stress factors within the tumour microenvironment like hypoxia and acidosis. Thus, during the later stages, the uncontrolled growth of tumors and the subsequent stress environment may stimulate the growth of metastases, proliferation and tumor angiogenesis. Understanding how they are expressed and regulated can help develop new ideas for pancreatic cancer prevention and treatment, as well as aid in understanding the pathophysiology of the disease.

V. Mechanisms and Receptors of T-Cell Exhaustion in Pancreatic Cancer

Chemotherapy, radiotherapy and immunotherapy have shown little effect on PDAC excrescences and have a poor prognosis. The blocking of the delivery of the medicines, low response to treatment, reduced number of neoantigens in excrescences, weak immunogenicity of excrescences, acquired natural resistance to therapies, genetic and epigenetic instability, and the special immunosuppressive excrescence medium (TME) were proposed as the factors that impair the delivery of the medicines and render the low treatment response. The very complex pancreatic TME regulates infiltration of immunosuppressive cells and the activity of susceptible regulatory molecules (Figure 1) which causes the dysfunction or downregulation of the antitumor immune response, including T cell prostration.[50]

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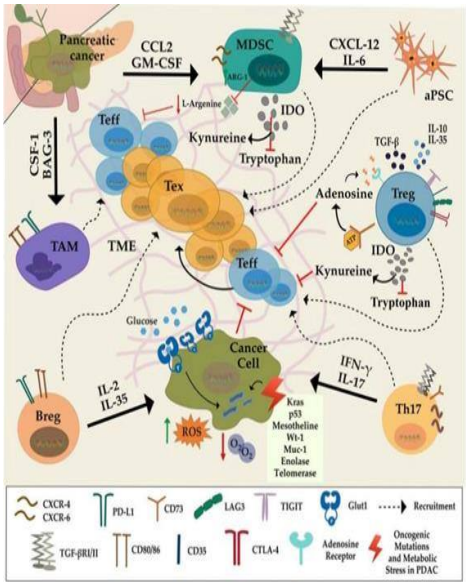


Figure 1. **T-cell exhaustion in pancreatic cancer: cellular and molecular immunomodulatory factors**

Myeloid-derived suppressor cells (MDSCs) and tumor associated macrophages (TAMs) exert direct and indirect T-cell suppression, such as by the tumor produced proteins, including granulocyte-macrophage colony stimulating factor (GM-CSF), C-C motif chemokine ligand 2 (CCL2), colony stimulating factor 1 (CSF1), and Bcl2-associated athanogene 3 (BAG3). Activated pancreatic stellate cells (aPSCs) suppress antitumor immune cells, and maintain suppressive cells in the stroma. In addition they activate immune checkpoints on T cells by the secretion of interleukin 6 (IL-6) via a C-X-C Motif chemokine 12 (CXCL12)-dependent pathway. Moreover, they stimulate the expansion of MDSCs and IL-35-secreting regulatory B cells (Bregs). Intratumoral

Tregs suppress the proliferation of Teffs and cause dysfunction of Teffs by secreting inhibitory cytokines such as TGF-β, IL-10, and IL-35. Tregs also produce indoleamine 2,3-dioxygenase (IDO) to boost kynurenine levels and deplete tryptophan in the tumor microenvironment (TME) that is needed for effector T cells to function properly. T cells are impaired in the TME by decreased levels of L-arginine, which are associated with decreased anti-tumoral responses. Th17 cells antagonize the activity of Tregs, but the function of the IL-17 secreted by them is controversial. In addition, mutations in KRAS, enolase and mesothelin in the TME alter T cell checkpoints and drive T cell exhaustion. Upregulation of GLUT-1 gene expression by Kirsten Rat Sarcoma (KRAS) oncogene is responsible to the Warburg effect. This mitochondrial dysfunction results in increased amounts of reactive oxygen species (ROS) within pancreatic cancer cells, which allows cells to progress at a faster rate.

Viral immunologists first described a unique state of T cells, called T-cell exhaustion, that develops during chronic antigen stimulation, such as that from continuous exposure to infectious agents, that triggers T-cell receptor (TCR) signaling during chronic infections, and that is more common in older individuals. There is now a growing body of evidence that T cells that somehow survive the initial activation process develop into two types of cells:

1. Progenitor or memory-like T cells.
2. Terminally differentiated exhausted T cells (Tex).

However, in oncological settings, certain stimuli, TME components, tumor type, and antigen presentation drive the development of the unique immunophenotypic and molecular properties of Tex. Current studies and studies for therapeutic reactivation indicate the possibility of a treatment window so exhaustion is a condition that can develop and has a progression of dysfunctional stages. To target solid tumors, such as pancreatic ductal adenocarcinoma (PDAC), that are resistant to immunotherapy, it is important to understand the signalling pathways that give rise to exhaustion. Therefore, we highlight recent advances regarding epigenetic and transcriptional changes to T-cell exhaustion upon receipt of tumour-reactive immunomodulatory signals from the pancreatic tumour microenvironment. Moreover, we discuss relevant clinical studies and new targets in PDAC immunotherapy.

VI. Metabolic Changes in T-Cell Exhaustion

The metabolic changes in T cells may be viewed as a response to the increased energy requirement during activation of T cells. Furthermore, these changes are essential for the differentiation and functional efficiency of T cells that are highly dependent on energy metabolism by-products. Naive and resting cells produce ATP mainly by oxidative phosphorylation[60] and both cancer cells and T cells undergo aerobic glycolysis. Depression of aerobic glycolysis significantly reduces effector cytokine production in activated T cells. [61]

In the early stages of exhaustion, the amount of glucose transport is limited by the downregulation of GLUT1 and excessive glucose consumption by the tumor cells.[60,61] In chronic infections, the Tex cells have increased number of dysfunctional mitochondria, leading to increased generation of reactive oxygen species (ROS) and reduced mitochondrial function. In contrast to metabolic changes during exhaustion in infectious environments, TILs show decreased mitochondrial function and mass. But the mitochondrial mass of TILs is not affected by PD-1 blockade despite successful tumor regression under PD-1 blockade. Hypoxia is a known factor that contributes to the growth and survival of solid tumors. T cell activity in hypoxia is regulated in part by hypoxia-inducible factor (HIFα) and von Hippel-Lindau (Vhl), a tumor suppressor and a negative regulator

of HIF, respectively. HIFα activity seems to have a profound effect on the transcriptional state and differentiation of T cells, given that the simultaneous occurrence of tumors and infections is deeply affected by increased HIFα activity.

VII. Clinical presentation/ signs and symptoms

7.1 Symptoms:These are all due to compression of the duodenum, the pancreatic duct, the celiac and mesenteric nerves, and the bile duct, and ultimately lead the patient to seek medical attention, so pancreatic cancer can be detected rather quickly. When the tumor in the head of the pancreas is very small, the only symptom may be mild jaundice. Often there is a previous period of abdominal or back pain before the onset of obstructive jaundice. Other signs of warning consist of malabsorption and sudden start of diabetes mellitus. If the pancreatic duct is very obstructed, a pancreas tumour can also lead to pancreatitis, which is a less common symptom, and must therefore be suspected if this symptom is encountered in a patient with no known risk factors. Tumors of the tail of the pancreas may have early complaints which are related directly to the primary tumor, such as pain in the left upper quadrant or flank of the abdomen, but more often present with systemic symptoms due to metastatic disease.

7.2 Sign: A large-scale case-control study identified twelve primary warning signs associated with pancreatic cancer, which include weight loss, abdominal pain, nausea and vomiting, bloating, dyspepsia, new-onset diabetes, changes in bowel movements, pruritus, fatigue, back pain, shoulder pain, and jaundice.[67] Specific features strongly linked to pancreatic tumors included back pain (Odds Ratio (OR): 1.33, 95% CI: 1.18–1.49), fatigue (OR: 1.42, 95% CI: 1.25–1.62), and new-onset diabetes (OR: 2.46, 95% CI: 2.16–2.80).[67] Furthermore, five specific symptoms—back pain, shoulder pain, dysphagia, altered bowel habits, and fatigue—were shown to persist for more than six months prior to formal diagnosis.[68] Regarding mood disorders, a thorough evaluation revealed that depressive symptoms acted as the primary initial presentation in approximately 38% to 45% of pancreatic cancer cases. Furthermore, a modern systematic review listed nine different clinical presentations that are typical of advanced pancreatic cancer[69,70]. Among them, upper abdominal pain and hyperglycemia have been observed as the result of the neoplastic nerve infiltration and is often present for >6 months before an early clinical diagnosis is considered by various investigators[72-82].

VIII. Diagnostic investigations

Diagnosis relies mainly on sophisticated imaging modalities like computed tomography (CT), magnetic resonance imaging (MRI), positron emission tomography (PET) and endoscopic ultrasound (EUS) along with

tissue procurement techniques. The only possible way to detect small tumors and to perform a curative surgery is to achieve early detection. Here we describe the different available diagnostic methods, discuss the evidence for their clinical use, and compare their merits and drawbacks. These findings are summarized in the table below and the figure below suggests an evidence-based diagnostic algorithm for the evaluation of patients suspected of having pancreatic cancer.

Diagnostic modalities	**Advantages**	**Limitations**
MDCT	Most widely available; Best validated; Most cost-effective	Risk of nephrotoxicity; Exposure to radiation
MRI	Superior tissue imaging; Better visualization of local pancreatic disease; Avoids iodine and radiation	High cost; Limited accessibility; Contraindicated in patients with specific metal implants
EUS +/- FNA	Safe and minimally invasive; Excellent sensitivity; Capable of detecting small tumors; Allows for histological sampling	Limited availability in certain regions; Highly operator-dependent; Cannot reliably detect distant metastases
PET/CT	Effective for detecting metastatic disease; Clarifies ambiguous CT results; Useful for monitoring treatment response and recurrence	High cost; Less widely available; Involves exposure to radiation and contrast agents

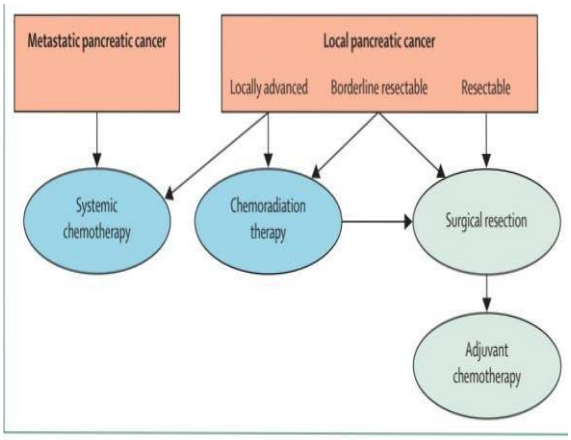
CT: Computed tomography; MDCT: Multi-detector computed tomography; MRI: Magnetic resonance imaging; EUS: Endoscopic ultrasound; FNA: Fine needle aspiration; PET: Positron emission tomography.

Table 1: Benefits and limitations of pancreatic cancer diagnostic modalities

IX. Treatment

Radiation treatment, chemotherapy, surgery, and palliative care are used to treat pancreatic cancer. The treatment options in a multidisciplinary approach are dependent on the stage of the pancreatic cancer (figure 2). Treatment of IPMN and MCN is discussed. [74]

Figure2: Schematic algorithm of treatment for pancreatic cancer



IX.1 Surgery

Suggestion for surgical resection

Surgical resection is considered the first-line curative therapy, and the survival is much longer than other therapies. In pancreatic cancer, the extent of the primary tumour is used to classify the condition into three types: resectable, borderline resectable, or locally advanced. However, the differences among these three classifications are not always clear-cut since resectability heavily depends on the surgical criteria applied. For example, an often-used definition of resectable pancreatic cancer is the absence of distant metastasis, and absence of tumor involvement of the superior mesenteric and portal veins, with clear gastroduodenal artery up to the hepatic artery and tumor abutment of the superior mesenteric artery < or = 180°.[75]

IX.2 Radiation therapy

Definitive chemoradiation treatment for locally advanced pancreatic cancer has been widely explored to understand its potential benefits. However, there is still some controversy over the benefits of chemoradiation therapy for survival. The ECOG 4201 study compared chemotherapy with chemoradiation in people with localized advanced pancreatic cancer. The median survival time was 9.2 months after gemcitabine-based chemotherapy and 11.1 months after chemoradiotherapy. In a separate randomized trial, however, researchers compared continuing chemotherapy to starting chemoradiation therapy after 4 months of gemcitabine, with or without erlotinib (LAP07 trial), but there were not enough patients enrolled to make definitive clinical

decisions. Additional studies are needed to confirm the intrinsic survival advantage of chemoradiation therapy for locally advanced pancreatic cancer.

IX.4 Palliative care

Chemoradiation is frequently used as the last resort in treating locally advanced pancreatic cancer. The basic advantages of this chemoradiation therapy have been extensively studied in patients with locally advanced disease. However, there are still questions about the survival improvement that chemoradiation therapy provides.[38] The research study ECOG 4201 was a comparison of standard chemotherapy and radiation for localized advanced pancreatic cancer. The median survival was 9.2 months in the gemcitabine-containing chemotherapy group and 11.1 months in the chemoradiotherapy group. Results from a randomized study after four months of gemcitabine plus or minus erlotinib (LAP07 trial) also showed that chemoradiation therapy did not provide a benefit over continuing standard chemotherapy in patients with locally advanced pancreatic cancer, however, further studies are needed to confirm the underlying survival benefits of chemoradiation therapy.

IX.5 Chemotherapy

Adjuvant chemotherapy for pancreatic cancer after surgical resection has demonstrated considerable clinical benefits. Adjuvant therapy (gemcitabine after surgery) is commonly used when the patient's physical well-being allows, and has been shown by the ESPAC-01 study to significantly improve survival rates from pancreatic cancer that has been resected.[80] The CONKO-001 randomized study also found that six cycles of adjuvant gemcitabine therapy after surgery was associated with improved overall and disease-free survival compared with no adjuvant therapy following surgery.[81] In Japan, a randomized phase 3 study (JASPAC-01) was performed to compare the efficacy of oral S-1 (fluoropyrimidine derivative) to gemcitabine as a postoperative adjuvant therapy.[82] The 2-year overall survival (OS) rate was 70% for S-1 and 53% for gemcitabine. This led to a conclusion that S-1 is superior to gemcitabine (hazard ratio 0.56, 95% CI 0.42–0.78; $p<0.0001$), which led to a change in the standard protocol for adjuvant therapy in Japan. But in order to be widely adopted around the world, these results require correct validation in other countries. Various randomized trials to assess upfront surgery versus different neoadjuvant chemotherapy or chemoradiation regimens are ongoing around the world. The results of a phase 3 trial of neoadjuvant chemotherapy with gemcitabine and S-1 before surgery versus immediate open surgery for resectable pancreatic cancer (JSAP-05/Prep-02) are expected to give definitive answers to the question of overall survival.

Chemotherapy is an essential component of the first-line treatment of metastatic pancreatic cancer. In a landmark clinical trial conducted in 1997, fluorouracil was compared to gemcitabine, with the median survival in the fluorouracil control group being 4.41 months, and the gemcitabine chemotherapy group showing a median survival of 5.65 months.[84]

Since then, gemcitabine has been a cornerstone of chemotherapy and several clinical studies have been conducted to evaluate new treatment guidelines in comparison to gemcitabine alone. A post hoc analysis indicated that there was a greater survival benefit in patients who developed grade 2 skin rash or developed progression after erlotinib administration, compared with those who did not.[85] So the ongoing application of gemcitabine with erlotinib chemotherapy is mainly recommended for those that develop skin rashes after treatment. The phase 3 ACCORD-11 trial tested the FOLFIRINOX regimen (oxaliplatin 85 mg/m², folic acid/leucovorin 400 mg/m², irinotecan 180 mg/m², and fluorouracil 400 mg/m2 followed by a continuous infusion over 46 hours every 14 days). In metastatic cases, the FOLFIRINOX protocol was more effective than Gemcitabine in terms of response, progression-free survival and overall survival. This is despite the absence of convincing, good evidence in favor of the high effectiveness of alternative-line chemotherapy, and may be beneficial for a patient who continues to have a good performance status after first-line chemotherapy, for metastatic pancreatic cancer.

Drug Approved for Pancreatic Cancer

- Capecitabine (Xeloda)
- Fluorouracil (5-FU)
- Gemcitabine (Gemzar)
- Irinotecan (Camptosar)
- Leucovorin (Wellcovorin)
- Nab-paclitaxel (Abraxane)
- Nanoliposomal irinotecan (Onivyde)
- Oxaliplatin (Eloxatin)

X. Pancreatic cancer is highly resistant to Chemotherapy drugs

In 1997, FDA approved gemcitabine as a first-line chemotherapy drug for people with locally advanced or metastatic pancreatic cancer. This approval was based on a randomized Phase III clinical study which initially

involved 126 patients, with better clinical outcomes, as well as an increase in median survival, seen in the group receiving gemcitabine as opposed to those receiving 5-fluorouracil (5-FU). Clinical benefit response rate was 23.8% for patients treated with gemcitabine, and 4.8% for patients treated with 5-FU ($p = 0.0022$). Patients treated with gemcitabine had a median survival time of 5.65 months vs 4.41 months with 5-FU ($p = 0.0025$). In the past few decades, many efforts have been undertaken to improve the clinical efficacy of chemotherapy schemes, and one such effort was the use of gemcitabine instead of 5-FU. Still, gemcitabine remains the recommended treatment for patients with pancreatic cancer. Although a number of clinical trials have evaluated gemcitabine combined with other cytotoxic agents (5-FU, cisplatin, oxaliplatin and capecitabine)[88–91] or biological/targeted agents (erlotinib, cetuximab and bevacizumab)[92–94] for some of these combination therapies, improved clinical responses and relatively longer progression-free intervals have been detected, but none have demonstrated significant superiority over gemcitabine monotherapy as first-line treatment. There is not much difference in overall survival rates. One other possible cause of this extreme medication resistance is the severely compromised drug supply mechanism frequently noticed in pancreatic cancer.[95]

Pancreatic tumors are also very dense solid tumors, where the chemotherapeutic agents have very little ability to penetrate and get into the stromal tissue areas because of the intense desmoplastic reaction. In addition, there is evidence that chemotherapy can paradoxically activate and promote intrinsic drug resistance mechanisms in the pancreatic cancer cell.

XI. Conclusion

Pancreatic cancer is still one of the most deadly and aggressive cancers. This can be a rapidly progressing and difficult-to-diagnose disease with resistance to conventional treatment. The molecular biology of this condition has made great strides and several reasons for high resistance have been found. These encompass KRAS oncogenesis, suppression of tumor suppressors, stromal interactions and a highly immunosuppressive tumor microenvironment. The broad desmoplasia, the sparsity of vessels, the metabolic changes and the high T-cell depletion in the tumor microenvironment limit the clinical benefits of gemcitabine-based therapies.

New and advanced approaches which may overcome this type of therapeutic resistance include cancer vaccines, checkpoint inhibitors in combination with chemotherapy, metabolic-targeting drugs and nanotechnology-based drug delivery. Targeted delivery systems, molecular profiling, and immunotherapy are the future methods for more successful and personalized treatments for pancreatic cancer. However,

substantial improvements in life expectancies and quality of life will require sustained research, constant innovation, and additional clinical trial development.

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XII. Appendix

Appendix A: List of Abbreviations

PDAC – Pancreatic Ductal Adenocarcinoma TME – Tumor Microenvironment CT – Computed Tomography MRI – Magnetic Resonance Imaging PET – Positron Emission Tomography EUS – Endoscopic Ultrasound FNA – Fine Needle Aspiration VEGF – Vascular Endothelial Growth Factor IL – Interleukin TNF – Tumor Necrosis Factor ROS – Reactive Oxygen Species GLUT-1 – Glucose Transporter 1 HIF – Hypoxia-Inducible Factor KRAS – Kirsten Rat Sarcoma Viral Oncogene TP53 – Tumor Protein p53 SMAD4 – Mothers Against Decapentaplegic Homolog 4

Appendix B: Approved Drugs for Pancreatic Cancer

Capecitabine (Xeloda) Fluorouracil (5-FU) Gemcitabine (Gemzar) Irinotecan (Camptosar) Leucovorin (Wellcovorin) Nab-paclitaxel (Abraxane) Oxaliplatin (Eloxatin)

Appendix C: Summary of Diagnostic Modalities

CT Scan – Used for initial diagnosis and staging MRI – Provides detailed soft tissue imaging PET Scan – Detects metastasis EUS – Helps detect small tumors and allows biopsy